

Smart City Management System

This system will manage various operations within a smart city, such as waste management, energy supply, water supply, public safety, citizen services, and analytics. The goal is to optimize services, improve citizen satisfaction, and automate administrative tasks.

Entities:

1. **Citizen:** Represents city residents, including registration and details.
 2. **Complaint:** Stores citizen complaints related to services (waste collection, energy, water, etc.).
 3. **WasteCollection:** Manages the schedules, routes, and completion status of waste collection.
 4. **EnergySupply:** Tracks energy distribution, service outages, consumption, and billing.
 5. **WaterSupply:** Manages water distribution, consumption, and billing.
 6. **Maintenance:** Tracks scheduled maintenance services for infrastructure.
 7. **Worker:** Represents city workers responsible for services like waste collection and maintenance.
 8. **Vehicle:** Tracks vehicles involved in services like waste collection or maintenance.
 9. **Contractor:** Represents external contractors for specific tasks like maintenance or repairs.
 10. **Incident:** Stores details of public safety incidents (accidents, hazards).
 11. **Report:** Manages reports generated from various services (e.g., usage, feedback).
 12. **Payment:** Records payments made for services like energy and water billing.
 13. **Notification:** Stores notifications sent to citizens for service updates, outages, or emergencies.
 14. **PublicService:** Represents different city services (Waste Management, Energy, Water Supply).
 15. **Department:** Represents city departments handling public services (Waste Department, Energy Department).
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Modules:

1. Waste Management:

- **Functionality:** Track waste collection schedules, routes, and manage citizen service requests related to waste collection.
- **Entities:** WasteCollection, Citizen, Worker, Vehicle, Complaint, Department, PublicService
- **Key Features:**
 - Scheduling waste collection for different areas.
 - Optimizing waste collection routes.
 - Citizens can request additional services or report issues (e.g., missed collections).
 - Workers assigned to specific routes, and vehicles are tracked for each route.

2. Energy Management:

- **Functionality:** Monitor energy usage, report outages, and manage billing for city residents.

- **Entities:** EnergySupply, Citizen, Complaint, Payment, Department, PublicService
- **Key Features:**
 - Track energy consumption for each registered household.
 - Generate and manage energy bills.
 - Handle citizen complaints regarding energy outages.
 - Analyze energy consumption patterns and predict demand.

3. Water Supply Management:

- **Functionality:** Manage water distribution and consumption, track usage, and handle billing.
- **Entities:** WaterSupply, Citizen, Payment, Complaint, Department, PublicService
- **Key Features:**
 - Monitor water consumption and generate bills.
 - Handle citizen complaints related to water supply (e.g., low pressure, supply issues).
 - Analyze water usage trends for efficient management.

4. Public Safety:

- **Functionality:** Citizens can report incidents (accidents, safety hazards) and track the resolution process.
- **Entities:** Incident, Citizen, Worker, Contractor, Complaint, Department, PublicService
- **Key Features:**
 - Allow citizens to report public safety incidents.
 - Assign workers or contractors to resolve incidents.
 - Track the status and completion of each incident report.
 - Generate reports on the frequency and types of incidents.

5. Service Scheduling:

- **Functionality:** Manage schedules for maintenance workers and contractors, and track their task completion status.
- **Entities:** Maintenance, Worker, Contractor, Vehicle, Department
- **Key Features:**
 - Assign maintenance or inspection tasks to workers or contractors.
 - Track the schedules and tasks assigned to each worker.
 - Monitor the availability and task history of city vehicles involved in maintenance.

6. Citizen Management:

- **Functionality:** Handle citizen registration, manage complaints, and send notifications.
- **Entities:** Citizen, Complaint, Notification, PublicService
- **Key Features:**
 - Citizens can register and update their details in the system.

- Complaints regarding public services can be lodged and tracked.
- Send notifications to citizens regarding service outages, updates, or emergencies.

7. Analytics Module:

- **Functionality:** Generate reports and insights on city services like energy consumption, water usage, waste disposal, and citizen feedback.
 - **Entities:** Report, PublicService, Department, Citizen, Complaint
 - **Key Features:**
 - Analyze data related to energy usage, water supply, and waste management.
 - Generate insights on citizen feedback, complaint resolutions, and service performance.
 - Provide department-level analytics to improve service efficiency and planning.
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Detailed Modules Description:

1. Waste Management

- **Description:** This module manages the waste collection processes within the smart city. It tracks waste collection schedules, assigns workers to specific routes, and ensures that vehicles are dispatched for waste pickup. It also handles citizen service requests related to waste collection.
 - **Key Functions:**
 - Assign workers and vehicles to waste collection routes.
 - Maintain and track waste collection schedules.
 - Record and resolve citizen waste-related complaints.
 - Manage vehicle usage for waste collection.
 - **Entities Involved:**
 - WasteCollection, Worker, Vehicle, Complaint
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2. Energy Management

- **Description:** This module handles the monitoring and billing of energy usage across the city. It tracks energy consumption by citizens, manages billing processes, and logs any service outages. Citizens can view their energy usage and make payments through this module.
 - **Key Functions:**
 - Monitor energy consumption for each citizen.
 - Generate and manage bills based on energy usage.
 - Log energy outages and incidents.
 - Provide citizens with the ability to view energy usage and pay bills.
 - **Entities Involved:**
 - EnergySupply, Citizen, Payment, Incident
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3. Water Supply Management

- **Description:** This module ensures the smooth distribution of water across the city and tracks water consumption by citizens. It also manages billing for water usage and records any complaints related to water supply, such as leakages or disruptions.
- **Key Functions:**
 - Track and monitor water distribution and consumption.
 - Generate bills based on water usage.
 - Manage citizen complaints related to water supply issues.
 - Log and resolve water supply outages or incidents.
- **Entities Involved:**
 - WaterSupply, Citizen, Complaint, Incident, Payment

4. Public Safety

- **Description:** This module allows citizens to report safety-related incidents, such as accidents, crime, or hazardous conditions. The system tracks the resolution of these complaints and manages the workforce assigned to address the incidents.
 - **Key Functions:**
 - Citizens can report safety-related incidents.
 - Assign workers to resolve public safety complaints.
 - Track the status and resolution of incidents.
 - Notify citizens of the status of their complaints.
 - **Entities Involved**:**
 - Incident, Worker, Citizen, Complaint, Notification
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5. Service Scheduling

- **Description:** This module manages the scheduling of maintenance tasks and public service activities. It tracks the schedules of workers and contractors and ensures that maintenance tasks like repairing infrastructure or addressing complaints are handled promptly.
 - **Key Functions:**
 - Schedule maintenance tasks and services.
 - Assign contractors and workers to specific jobs.
 - Track completion of maintenance tasks.
 - Manage the availability of workers and contractors.
 - **Entities Involved:**
 - Maintenance, Worker, Contractor, Department
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6. Citizen Management

- **Description:** This module focuses on the management of citizen information and their interactions with the city's services. It handles citizen registration, complaint management, and notifications related to various services like billing, maintenance, or public safety.
 - **Key Functions:**
 - Register citizens in the system.
 - Handle citizen complaints related to public services.
 - Send notifications to citizens regarding service updates.
 - Track citizen usage of public services like energy, water, and waste management.
 - **Entities Involved:**
 - Citizen, Complaint, Notification
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7. Analytics Module

- **Description:** The analytics module provides data-driven insights into the city's operations. It generates reports based on data from energy consumption, water supply, waste management, and citizen feedback. The analytics help the administration in decision-making and improving service efficiency.
 - **Key Functions:**
 - Generate reports on energy usage, water consumption, waste collection, and incident resolutions.
 - Provide insights based on citizen feedback.
 - Analyze data for performance tracking of services.
 - Assist in strategic planning by offering data insights for city officials.
 - **Entities Involved:**
 - Report, Citizen, PublicService, Department
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Summary of Modules:

- **Waste Management:** Handles waste collection schedules, routes, and worker assignments.
- **Energy Management:** Monitors energy consumption and billing.
- **Water Supply Management:** Tracks water distribution, consumption, and billing.
- **Public Safety:** Reports incidents and tracks their resolution.
- **Service Scheduling:** Manages schedules for maintenance workers and contractors.
- **Citizen Management:** Handles citizen registration, complaints, and notifications.
- **Analytics Module:** Provides data insights on energy/water usage, waste collection, and public feedback.